

## USECASE 2 IHL

RAG:

```
# =====
```

```
# SIMPLE RAG CHATBOT USING QWEN LLM
```

```
# Example: College Student Information
```

```
# =====
```

```
!pip install -q transformers accelerate sentence-transformers
```

```
from transformers import pipeline
```

```
from sentence_transformers import SentenceTransformer
```

```
import torch
```

```
# Load Qwen LLM
```

```
chatbot = pipeline(
```

```
    "text-generation",
```

```
    model="Qwen/Qwen2.5-0.5B-Instruct",
```

```
    device=0
```

```
)
```

```
chatbot.model.generation_config.max_length = None
```

```
# Knowledge Base
```

```
documents = [
```

```
    """
```

```
    Our college offers undergraduate programs in Computer Science,
```

Artificial Intelligence and Data Science, Electronics and Communication,  
and Mechanical Engineering.

""",

"""

The Computer Science department teaches programming languages,  
data structures, algorithms, databases, computer networks,  
operating systems, and artificial intelligence.

""",

"""

Students can use the college library to access textbooks,  
reference books, journals, and digital learning resources.

""",

"""

The college provides laboratories for programming, computer networks,  
electronics, and other practical learning activities.

""",

"""

Students can participate in technical events, workshops,  
hackathons, coding competitions, and project activities.

"""

]

```
print("Knowledge base created!")
```

```
# Create embedding model
```

```
embedding_model = SentenceTransformer(  
    "sentence-transformers/all-MiniLM-L6-v2"  
)
```

```
# Convert documents into embeddings
```

```
document_embeddings = embedding_model.encode(  
    documents,  
    convert_to_tensor=True  
)
```

```
print("Document embeddings created!")
```

```
# Retrieval function
```

```
def retrieve_documents(query, top_k=2):
```

```
    query_embedding = embedding_model.encode(  
        query,  
        convert_to_tensor=True  
    )
```

```
    scores = torch.nn.functional.cosine_similarity(  
        query_embedding,  
        document_embeddings
```

```
)
```

```
top_results = torch.topk(scores, k=top_k)
```

```
retrieved_docs = [  
    documents[i] for i in top_results.indices  
]
```

```
return retrieved_docs
```

```
# Start RAG chatbot
```

```
print("\n🤖 RAG Chatbot Started!")
```

```
print("Type 'exit' to stop.\n")
```

```
while True:
```

```
    user_input = input("You: ")
```

```
    if user_input.lower() == "exit":
```

```
        print("Bot: Goodbye!")
```

```
        break
```

```
# Retrieve relevant information
```

```
retrieved_docs = retrieve_documents(user_input)
```

```
# Combine retrieved information
```

```
context = "\n".join(retrieved_docs)
```

```
# Create prompt
```

```
prompt = f"""
```

You are a helpful college information assistant.

Answer the user's question using only the information provided below.

If the answer is not available in the knowledge base, say:

"Sorry, this information is not available in my knowledge base."

Knowledge Base:

```
{context}
```

User Question:

```
{user_input}
```

```
"""
```

```
# Generate answer using Qwen
```

```
response = chatbot(
```

```
    prompt,
```

```
    max_new_tokens=50,
```

```
    do_sample=True,
```

```
    temperature=0.7
```

```
)
```

```
answer = response[0]["generated_text"]
```

```
print("Bot:", answer)
```

```
print()
```

INPUT/OUTPUT:

```
RAG Chatbot Started!
... Type 'exit' to stop.

You: WHAT SUBJECTS ARE TAUGHT IN COMPUTER SCIENCE?
[transformers] Both `max_new_tokens` (=50) and `max_length` (=20) seem to have been set. `max_new_token
Bot:
You are a helpful college information assistant.

Answer the user's question using only the information provided below.
If the answer is not available in the knowledge base, say:
"Sorry, this information is not available in my knowledge base."

Knowledge Base:

    The Computer Science department teaches programming languages,
    data structures, algorithms, databases, computer networks,
    operating systems, and artificial intelligence.

    The college provides laboratories for programming, computer networks,
    electronics, and other practical learning activities.

User Question:
WHAT SUBJECTS ARE TAUGHT IN COMPUTER SCIENCE?
The subject taught by the Computer Science department includes programming languages, data structures,

You: EXIT
```